

GPA Calculator.cpp

Project 1

Purpose

This program calculates a student's semester GPA. The user enters each course's credit amount followed by the letter grade they received. The program performs the calculations and then prints them out for the user in the terminal by showing the classes they have taken and their semester GPA. This is intended to be used by any Hunter College student as if it was part of the college's tools.

Inputs

The program receives the following inputs from the user:

- * Course credit amount
- * Letter grade for each course
- * Whether the user wants to enter another course
- * Whether the user wants to keep using the GPA calculator

The program accepts letter grades using uppercase or lowercase letters. For yes/no questions, the program accepts answers such as “yes”, “no”, “y”, and “n”, along with their capitalized version.

Outputs

The program displays the following outputs:

- * A welcome message
- * Instructions for using the GPA calculator
- * Examples of valid course input
- * Error messages for invalid input
- * The number of classes entered
- * The calculated cumulative GPA
- * A goodbye message when the program ends

Variables

Variable	Type	Purpose
valid_grades	string array	Stores the valid letter grades accepted by the program.
credit_points	double array	Stores the GPA point values that match the valid_grades array.
total_credits	double	Stores the total number of valid course credits entered.
total_points	double	Stores the total GPA points earned from all valid courses.
gpa	double	Stores the final calculated GPA.
class_counter	int	Counts the number of valid classes entered.
answer	string	Stores the user's answer about whether to keep using the calculator.
running	bool	Controls whether the full GPA calculator keeps running.
user_course_credit	double	Stores the credit amount entered by the user for one course.
user_letter_grade	string	Stores the letter grade entered by the user for one course.
grade_point	double	Stores the GPA point value for the user's current letter grade.
valid_course	bool	Controls whether the current course input is valid.
ask_another_course	string	Stores the user's answer about whether to enter another course.

grade_found	bool	Tracks whether the user's letter grade was found in the valid_grades array.
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Key Design Choices

* **Arrays:** The program uses arrays to connect valid letter grades with their GPA point values. This avoids writing many separate if-statements for each grade.

* **The while-loops:** The program uses three while-loops. First one allows the user to do multiple GPA calculations without exiting the program from the terminal. Second one allows the user to add more courses for the GPA calculator to add. Finally, the third one validates the user inputs and repeats as long as the user does not enter a valid course grade or credit.

* **<cctype> library and case-insensitivity:** The program converts letter grades to uppercase so grades like b, a+, and wu can be handled the same way as B, A+ and WU. To do so, the library <cctype> is used along with the function toupper() that will convert any string to a capitalized form.

* **Valid answers:** The program converts yes/no answers to lowercase so answers like “YES” or “NO” can be handle consistently with “yes” and “no” by using the function tolower() from the <cctype> library. Likewise, it allows the user to respond with only Y or N or their lowercase forms.

* **Variable reset:** Although the variables are declared and initialized at the top of the program, they are called throughout the code, at the beginning of each while-loop, so that their values always return to their initialized value with each loop-repetition done by the user. By doing so, old data stored in the variables cannot affect the next result. In addition, the program also resets course validation variables when a new course begins so previous valid input does not affect the next course.

* **User information storage:** Since each iteration erases the information previously entered by the user so that it cannot create a conflict with the new information of credits and grades, the loop immediately stores the information temporarily held in user_course_credit, user_letter_grade and grade_points into total_points and total_credits, variables that not only hold the user information, but also uses it to perform the necessary multiplication and addition that will be used later on. Therefore, although the user information is essentially deleted, said information serves its purpose for the algebra and the results are stored safely to be later used with no conflicts.

* **Division by zero:** The program checks whether `total_credits` are greater than 0 before dividing to avoid division by zero. Although unlikely by the exclusion of 0 as valid input, this can be helpful for future modifications or to avoid user's potential chances to break the code.

Program Steps (Algorithm)

1. Display the welcome message, instructions, valid grades, and examples.
2. Start the main loop while the user wants to keep using the calculator.
3. Reset GPA calculation variables for a brand-new start.
4. Ask the user to enter course credits and a letter grade.
5. If the credit input is not numeric, or if it is out of the 1 to 9 parameter, clear the input error and ask again.
6. Compare the user's letter grade with the `valid_grades` array.
7. If the grade is found, store the matching GPA point value from `credit_points`. If the grade is not found, display an error message and ask again.
8. Add the valid course's GPA points to `total_points` and the valid course's credits to `total_credits` while also increasing `class_counter` by 1.
9. Ask whether the user wants to enter another course.
10. If the user says yes, they can enter another course and repeat the course input process.
11. If the user does not enter another course, calculate GPA by dividing `total_points` by `total_credits`.
12. Display the number of classes entered and the cumulative GPA.
13. Ask whether the user wants to keep using the GPA calculator.
14. If the user answers yes, start another GPA calculation. If the user answers no, end the program and display the goodbye message.

Functions

Function: `main()`

Purpose: Runs the full GPA calculator. It displays instructions, receives user input, validates course credits and letter grades, calculates GPA, displays the result, and allows the user to run another calculation.

Sample Input/Output

Output	Input
<pre>===== Welcome to Hunter College GPA Calculator ===== This program calculates your semester GPA. How to use: 1. Enter the number of credits followed by the letter grade. 2. You may enter the input with or without a space, such as 3A or 3 A. 3. Valid grades are: A+, A, A-, B+, B, B-, C+, C, D, F, WU, FIN, FAB. 4. Credit values must be greater than 0 and less than 10. Examples: 3A = 3 credit course with an A grade 4b+ = 4 credit course with a B+ grade 2wu = 2 credit course with a WU grade</pre>	
Enter credits and letter grade:	3A+
Do you wish to enter another course? [yes/no or y/n]:	y
Enter credits and letter grade:	4A-
Do you wish to enter another course? [yes/no or y/n]:	yes
Enter credits and letter grade:	2.5 A
Do you wish to enter another course? [yes/no or y/n]:	Yes
Enter credits and letter grade:	1B+
Do you wish to enter another course? [yes/no or y/n]:	NO
<pre>===== Classes taken: 4 Cumulative GPA: 3.82 =====</pre>	
Do you want to keep using the GPA Calculator? [yes/no or y/n]:	No
Thank you for using the GPA Calculator. Goodbye!	

Testing

Test Case 1: Perfect case scenario

Input:

3 A

no

n

Expected Result:

The program accepts the valid course, calculates the GPA, displays 1 class taken, displays a cumulative GPA of 4.00, and exits.

Test Case 2: 0 credits

Input:

0 A

3 A

no

n

Expected Result:

The program rejects 0 credits, asks for course input again, accepts 3 A, calculates the GPA, and exits.

Test Case 3: Invalid letter grade

Input:

3 Z

3 B+

no

n

Expected Result:

The program rejects Z as an invalid grade, asks for course input again, accepts 3 B+, calculates the GPA, and exits.

Test Case 4: Non-Numeric credit input

Input:

hello A

3 C

no

n

Expected Result:

The program rejects the non-numeric credit input, clears the input error, asks for course input again, accepts 3 C, calculates the GPA, and exits.

Test Case 5: Lower-case input

Input

3 a

Y

4 b+

No

N

Expected Result:

The program accepts lower-case grades, accepts Y and No as valid answers, calculates the GPA for 2 classes, and exits when the user enters N.